



**Strathfield Girls
High School**

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Student Number

Chemistry

2022
HIGHER SCHOOL CERTIFICATE
Trial Examination

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A formulae sheet, data sheets and Periodic Table are provided at the back of this paper and may be detached
- A Multiple-Choice Answer Sheet is provided at the back of this paper and should be detached

Total marks – 100

Section I – Pages 2–9 20 marks

- Attempt Questions 1–20
- Allow about 30 minutes for this section

Section II – Pages 10–24 80 marks

- Attempt Questions 21–32
- Allow about 2 hours and 30 minutes for this section

This paper MUST NOT be removed from the examination room

Section I

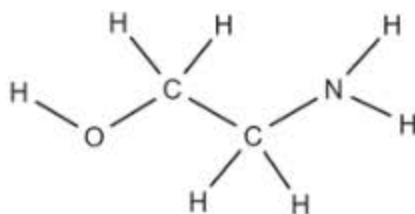
20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1 Which of the following substances is NOT a Brønsted-Lowry acid?
- (A) H_2PO_4^-
(B) NH_4^+
(C) H_2O
(D) BF_3
- 2 Which two compounds would be difficult to distinguish using infrared spectroscopy?
- (A) Ethanoic acid and ethanol
(B) Propanal and propanone
(C) Ethane and ethanamine
(D) Propene and propanamide
- 3 The structure of 2-aminoethanol is shown.



The mass spectrum of 2-aminoethanol contains a peak at m/z 31.

What is the most likely source of this peak?

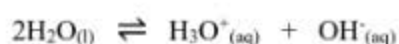
- (A) $[\text{CH}_2\text{NH}_3]$
(B) $[\text{NOH}]^+$
(C) $[\text{N}_2\text{H}_3]^+$
(D) $[\text{CH}_2\text{OH}]^+$

- 4 Equal volumes of two 0.01 mol L^{-1} solutions were mixed together.

Which pair of solutions would give the greatest mass of precipitate?

- (A) FeSO_4 and $\text{Ba}(\text{OH})_2$
(B) FeSO_4 and BaCl_2
(C) FeSO_4 and $\text{Ca}(\text{OH})_2$
(D) FeSO_4 and CaCl_2

- 5 The equation for the auto-ionisation of water is shown.



At 60°C , the water ionisation constant, K_w , is 9.55×10^{-14} .

What is the pH of water at 60°C ?

- (A) 7.14
(B) 7.00
(C) 6.92
(D) 6.51

- 6 Compound **W** shows two signals in its ^{13}C NMR spectrum.

Treatment of compound **W** with hot concentrated sulfuric acid produces compound **X**.

Compound **X** can be mixed with bromine water to produce compound **Y**.

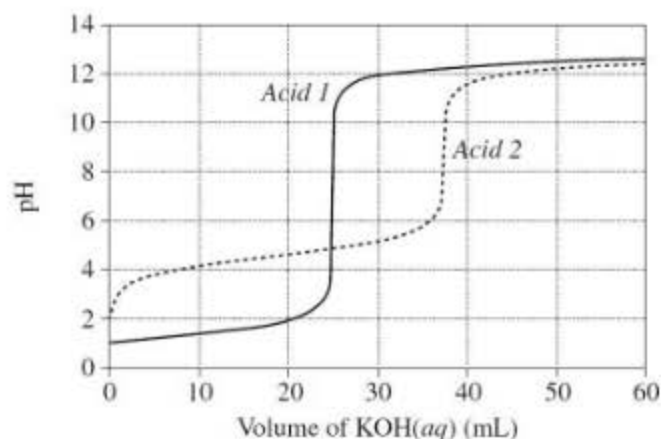
Compound **W** can be reacted with hot acidified potassium permanganate to produce compound **Z**.

Compound **Z** has one signal in its ^1H NMR spectrum.

Which of the following correctly identifies compounds **W**, **X**, **Y** and **Z**?

	Compound W	Compound X	Compound Y	Compound Z
(A)	ethanol	ethene	1,2-dibromoethane	ethanoic acid
(B)	butan-2-ol	but-2-ene	2,3-dibromobutane	butanone
(C)	propan-2-ol	propene	1,2-dibromopropane	propanone
(D)	propan-1-ol	propene	1,2-dibromopropane	propanoic acid

The graph shows changes in pH during the titrations of equal volumes of two acids, *Acid 1* and *Acid 2*. Use the graph to answer Questions 7 and 8.



7 Which indicator would be most suitable for BOTH titrations?

	Indicator	Colour Change Range (pH)
(A)	alizarin yellow R	10.0-12.0
(B)	phenolphthalein	8.3-10.0
(C)	bromothymol blue	6.0-7.6
(D)	4-nitrophenol	5.4-6.6

8 At which pH would the solution of the *Acid 2* titration be most effective as a buffer?

- (A) 4.5
- (B) 6
- (C) 8.5
- (D) 12

9 Which class of organic compound must contain at least two carbon atoms?

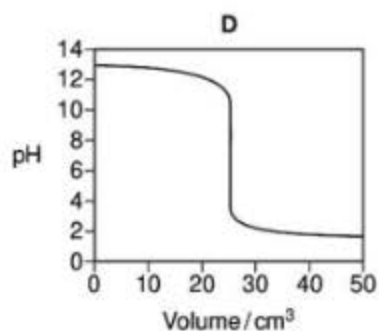
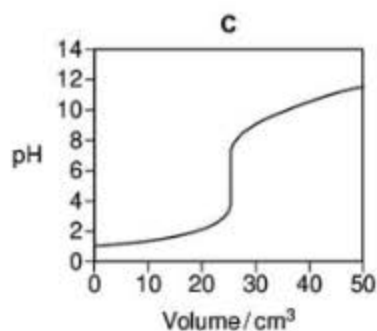
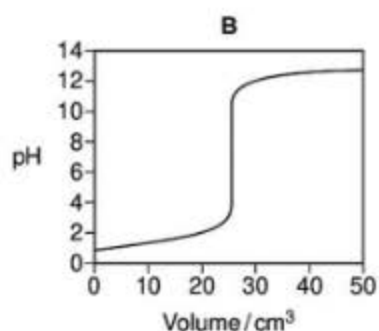
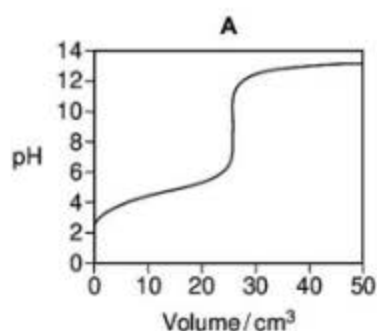
- (A) alkane
- (B) aldehyde
- (C) carboxylic acid
- (D) ester

- 10 The pK_a of hydrofluoric acid (HF) is 3.14. Oxalic acid ($H_2C_2O_4$) has a pK_a of 1.23.

The relative strength of each acid and its conjugate base can be identified as:

	Stronger Acid	Stronger Conjugate Base
(A)	HF	F^-
(B)	HF	$HC_2O_4^-$
(C)	$H_2C_2O_4$	$HC_2O_4^-$
(D)	$H_2C_2O_4$	F^-

- 11 Which of the following graphs would result from the titration of a *strong* acid with a *weak* base?



- (A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

- 12 Equal volumes of a monoprotic *strong* acid and a monoprotic *weak* acid, both with the same concentration, are titrated with the same base solution at 25°C.

Which of the following statements about the volume of base solution required to reach the equivalence point is correct?

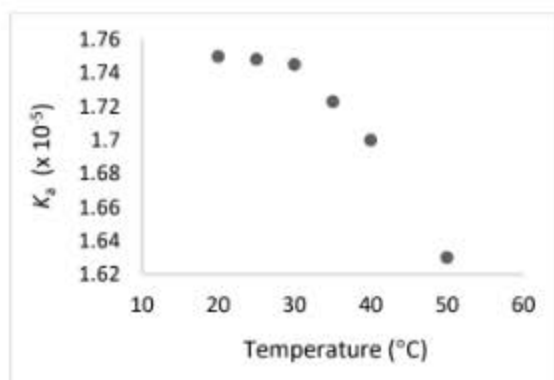
- (A) The strong acid will require more base solution than the weak acid
- (B) The weak acid will require more base solution than the strong acid
- (C) Both acid solutions will require the same amount of base solution
- (D) The amount of base solution required is determined by the acid's K_a at 25°C

- 13 Solid magnesium hydroxide is added to 150 mL of water to give a solution of pH 13.7.

What mass of magnesium hydroxide dissolved?

- (A) 2.19 g
- (B) 3.11 g
- (C) 4.38 g
- (D) 8.77 g

- 14 The K_a of ethanoic acid changes with temperature as shown in the graph.



The graph indicates that within the temperature range shown, as the temperature increases the acid becomes:

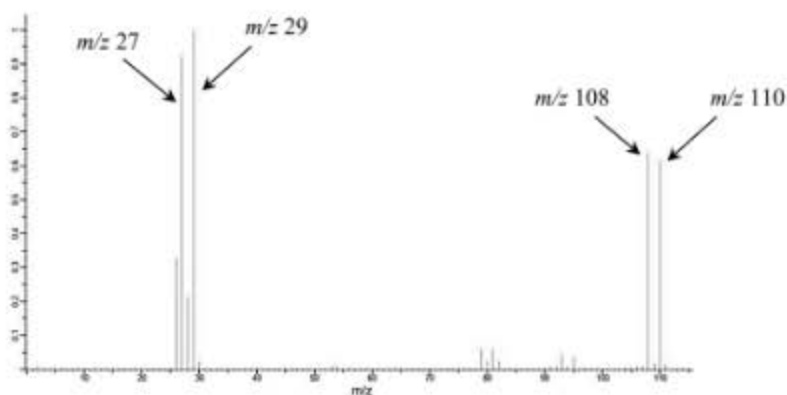
- (A) more ionised
- (B) less ionised
- (C) more concentrated
- (D) less concentrated

- 15 Which of the following compounds would be suitable for the preparation of a *primary standard* solution for use in an acid-base titration?
- (A) Hydrochloric acid
 (B) Sodium hydroxide
 (C) Sodium carbonate
 (D) Nitric acid
- 16 Equal volumes of $0.30 \text{ mol L}^{-1} \text{ AgNO}_3$ and $0.20 \text{ mol L}^{-1} \text{ Na}_3\text{PO}_4$ solutions are mixed together and a precipitate is observed.

Which option shows the concentration of all ions remaining in solution?

	$[\text{Ag}^+] \text{ mol L}^{-1}$	$[\text{NO}_3^-] \text{ mol L}^{-1}$	$[\text{Na}^+] \text{ mol L}^{-1}$	$[\text{PO}_4^{3-}] \text{ mol L}^{-1}$
(A)	0	0.15	0.30	0.05
(B)	0	0.30	0.60	0
(C)	0.15	0.15	0.30	0.10
(D)	0	0.15	0.30	0

- 17 The *mass spectrum* of an organic compound is shown.



Which compound was analysed?

- (A) 3-chlorobutan-1-ol
 (B) bromoethane
 (C) 1,3-dichloropropene
 (D) bromomethanol

- 18 A chemist used colorimetry to determine the concentration of chromium(VI) ions present in a river used for the discharge of industrial wastewater.

Four water samples were collected from the river. An appropriate ligand compound was added to 50 mL of each water sample and the total volume made up to 100 mL with distilled water.

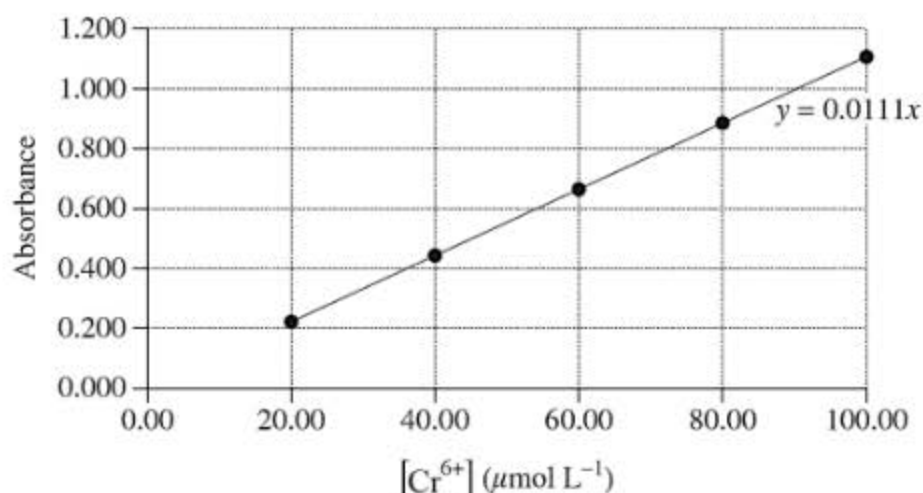
The absorbance of each prepared sample solution was measured. The results are shown in the table.

Sample	1	2	3	4
Absorbance	0.7625	0.7715	0.7695	0.7705

The chemist calculated the average absorbance using the results of all four samples.

She then prepared standard solutions of chromium(VI), measured the absorbance of each standard and constructed a calibration curve.

The calibration curve for chromium(VI) is shown.



What is the concentration of chromium(VI) in the river?

- (A) $3.6 \times 10^{-3} \text{ g L}^{-1}$
- (B) $7.2 \times 10^{-3} \text{ g L}^{-1}$
- (C) $8.2 \times 10^{-3} \text{ g L}^{-1}$
- (D) $8.8 \times 10^{-3} \text{ g L}^{-1}$

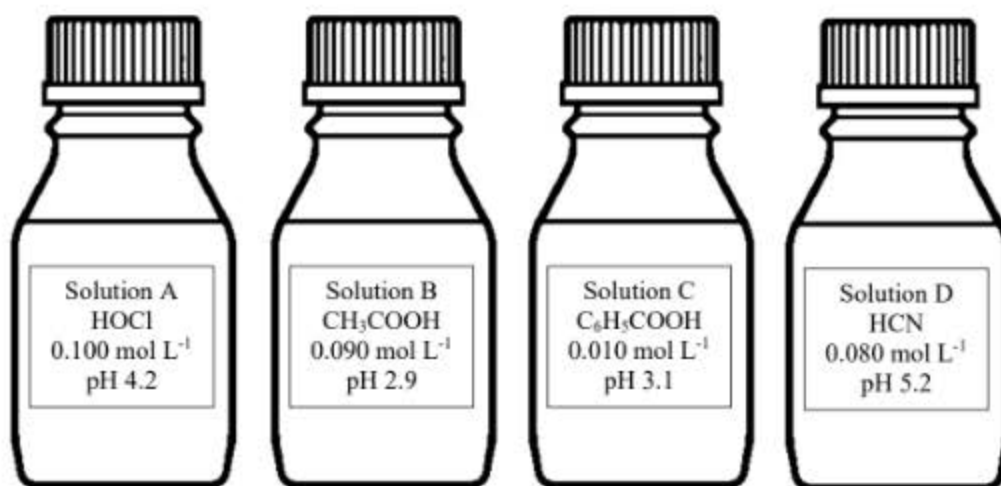
- 19 A chemist prepares 105 mL of 2.3 mol L^{-1} hydrochloric acid solution and 85 mL of 2.4 mol L^{-1} sodium hydroxide solution. The two solutions were at the same initial temperature before mixing.

The following information about the resulting solution is provided.

Mass of the final solution:	201 g
Temperature of the final solution:	34.0°C
Specific heat capacity of the final solution:	$3.99 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$
$\text{HCl}_{(\text{aq})} + \text{NaOH}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$	$\Delta H = -57 \text{ kJ mol}^{-1}$

Using the above information, the initial temperature of the two solutions would be:

- (A) 14.5°C
(B) 16.8°C
(C) 19.5°C
(D) 21.0°C
- 20 The diagram shows four labelled bottles, each containing 500 mL of an acid.



Which solution contains the strongest acid?

- (A) Solution A
(B) Solution B
(C) Solution C
(D) Solution D

Section II

80 marks

Attempt Questions 21–32

Allow about 2 hours and 30 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 25 and 26.

If you use this space, clearly indicate which question you are answering and indicate the use of this extra space with the question on the examination paper.

Question 21 (5 marks)

When ethanamine gas is bubbled into water, an ionisation reaction occurs.

- (a) Write a chemical equation for the ionisation of ethanamine in water.

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- (b) The pK_b of ethanamine is 3.29 at 25°C.

Calculate the pH of a 0.150 mol L⁻¹ aqueous solution of ethanamine.

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Question 22 (6 marks)

Hydrogen peroxide, a colourless liquid, is commonly used as an aqueous solution in bleach and antiseptic. As hydrogen peroxide decomposes into water and oxygen over a period of time, its concentration in commercial products has to be determined by a suitable titration.

Acidified potassium permanganate is often used in titrations to determine hydrogen peroxide concentration. The titration reaction is:



To perform the analysis, a chemist titrated 20.0 mL aliquots of the hydrogen peroxide solution against 0.0239 mol L⁻¹ acidified potassium permanganate solution. The results are as follows:

Titration Run	Titre (mL)
1	28.85
2	26.90
3	26.80
4	26.65
5	26.90

- (a) Given that no other chemical is used in the analysis, how would the equivalence point of each titration be recognised?

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- (b) Calculate the concentration of the hydrogen peroxide solution in g L⁻¹.

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Question 23 (11 marks)

A white powder was known to be a pure *ionic* substance. The diagram below gives the results of various tests used to identify this powder.



- (a) Identify an *anion* that could be in the white powder. Justify your choice by explaining the evidence from relevant test results.

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Question 23 continues on page 13

Question 23 (continued)

- (b) Identify a *cation* that could be in the white powder. Justify your choice by explaining the evidence from all relevant test results.

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- (c) Explain why it would be poor technique to use sodium sulfate in **step 5** instead of sulfuric acid.

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- (d) Assuming an excess of each indicated reagent is added at the various steps, identify all ions that could be present in the colourless solution obtained from **step 6**. Justify your answer.

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End of Question 23

Question 24 (7 marks)

A chemist prepares a *buffer* solution with propanoic acid and sodium propanoate.

- (a) Write a chemical equation to represent this buffer system as an equilibrium in aqueous solution. **1**

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- (b) Explain why this combination of chemicals will be an effective buffer. Refer to the equation written in part (a) to support your answer. **4**

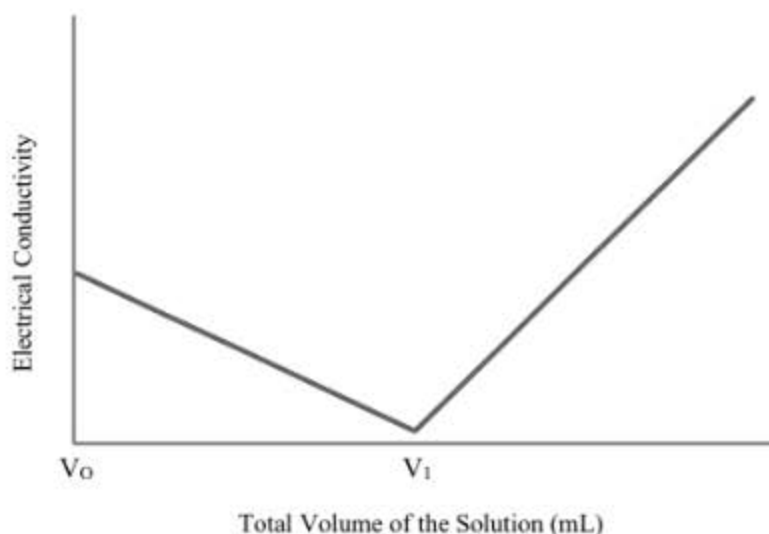
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- (c) Which TWO factors are important when *selecting* the weak acid and its conjugate base and *preparing* the buffer solution for a required pH? Justify your answer with reasons. **2**

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Question 25 (9 marks)

The graph shows the electrical conductivity of the resulting solution formed when 80 mL of 0.10 mol L^{-1} BaCl_2 is slowly added to 400 mL of 0.01 mol L^{-1} Ag_2SO_4 .



- (a) Write a net *ionic* equation for the chemical reaction that occurred when the BaCl_2 solution is added to the Ag_2SO_4 solution.

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- (b) Explain why the solution conducts electricity at volume V_0 but not significantly at volume V_1 .

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Question 25 continues on page 16

Question 25 (continued)

- (c) Determine the volume of BaCl_2 solution that has been added between volumes V_0 and V_1 .
Justify your answer through calculations.

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- (d) Explain why the electrical conductivity increases after volume V_1 .

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Question 26 (4 marks)

160 mL of 0.259 mol L^{-1} sulfuric acid is added to 185 mL of 0.695 mol L^{-1} potassium hydroxide.

Calculate the pH of the resulting solution.

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Question 27 (6 marks)

A chemist investigated the acid-base properties of three salts. She dissolved equal masses of sodium bicarbonate, ammonium ethanoate and potassium hydrogen sulfate in 20 mL volumes of distilled water. She measured the pH with a data logger and pH probe.

- (a) Complete the table to identify the acid-base property of the salt solutions measured.

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Name	Chemical Formula	Acidic, Basic or Neutral?
Sodium bicarbonate	NaHCO_3	
Ammonium ethanoate	$\text{NH}_4\text{CH}_3\text{COO}$	
Potassium hydrogen sulfate	KHSO_4	

- (b) Explain the acid-base property of each salt, using chemical equations to support your answer.

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- (c) Apart from failing to repeat the experiment, identify the major flaw in the chemist's method that could impact on the validity of her investigation.

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Question 28 (7 marks)

Cobalt and a ligand X react to form a coloured complex as shown in the general equation:



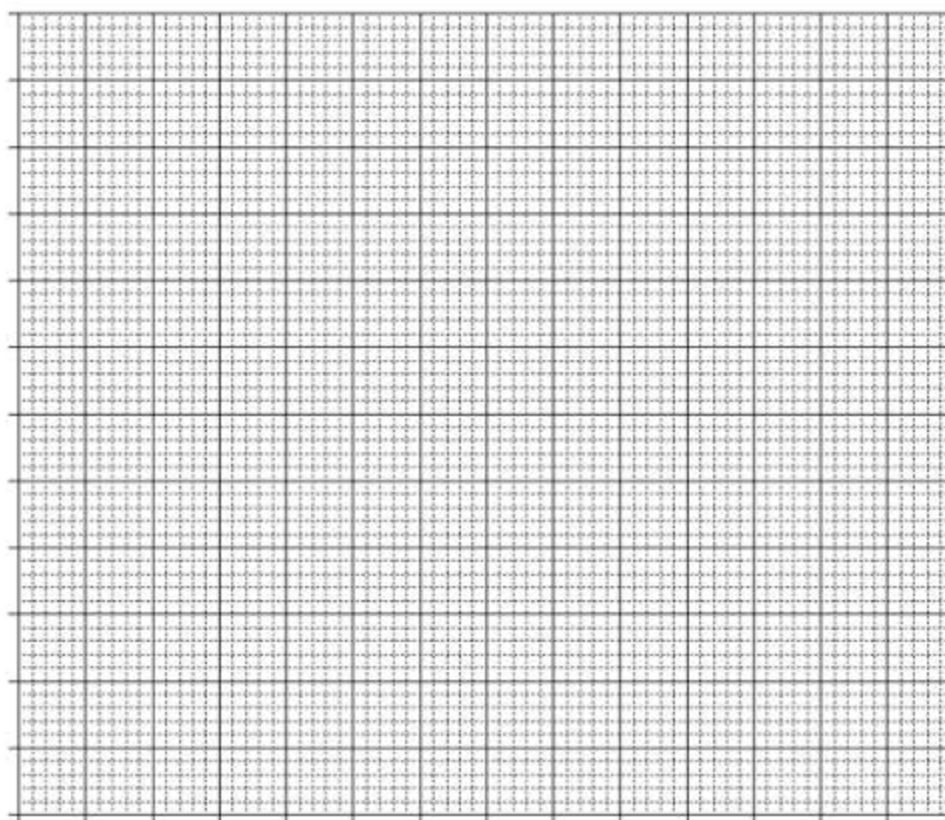
where a and b are numbers representing the ratio in which Co^{3+} and X combine.

To determine the stoichiometric ratio for this reaction, Co^{3+} and X solutions of equal concentration were mixed together in different ratios to make ten 70.0 mL samples. The intensity of colour in each sample was measured using a spectrophotometer. The results are shown in the table below.

	Samples									
Volume of Co^{3+} solution (mL)	0.00	3.00	6.00	9.00	14.0	22.0	35.0	48.0	61.0	70.0
Volume of X solution (mL)	70.0	67.0	64.0	61.0	56.0	48.0	35.0	22.0	9.00	0.00
Absorbance at 430 nm	0.00	0.34	0.66	1.02	1.07	0.92	0.67	0.42	0.17	0.00

- (a) Construct a graph of absorbance against the volume of Co^{3+} solution from 0.00 mL to 70.0 mL. Plot the data by drawing TWO lines of best fit.

4



Volume of Co^{3+} solution (mL)

Question 28 continues on page 19

Question 28 (continued)

- (b) Use the graph to determine the highest absorbance that would be obtained for a 70 mL sample, assuming unlimited combinations of the two solutions are measured.

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- (c) The reaction to form the coloured complex proceeds according to the general equation:



Determine the values for a and b . Justify your answer with reference to the graph in part (a).

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Question 29 (6 marks)

A chemist dissolved the calcium carbonate in a section of blocked pipe by soaking the section in 100.0 mL of 0.250 mol L⁻¹ hydrochloric acid. After the calcium carbonate had dissolved, the excess hydrochloric acid was titrated with 0.100 mol L⁻¹ sodium hydroxide, requiring 39.3 mL.

- (a) Write a chemical equation for the reaction between calcium carbonate and hydrochloric acid.

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- (b) Determine the mass of calcium carbonate that was in the blocked pipe.

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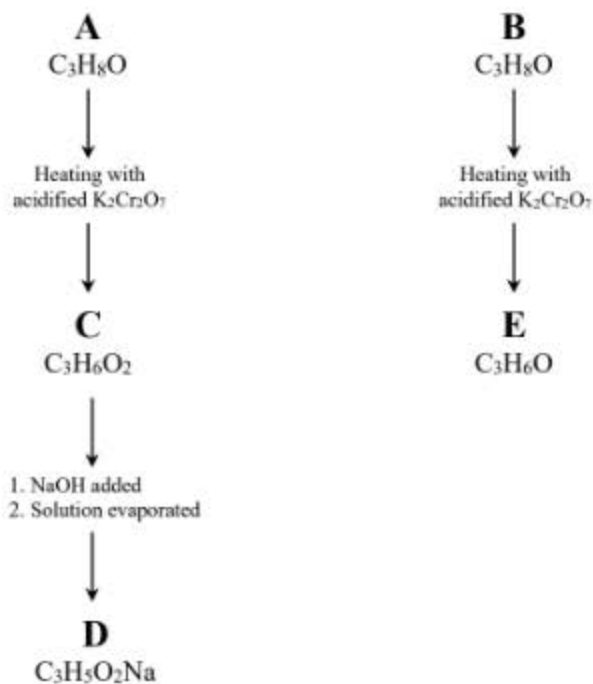
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Question 30 (8 marks)

Two different compounds, **A** and **B**, are *isomers* with the molecular formula C_3H_8O .

Compound A and **compound B** undergo a series of reactions as shown below.



- (a) Draw structural formulae for **compound C** and **compound E** in the space below.

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Compound C	Compound E
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- (b) Identify **compound A** and **compound B** by providing their systematic (IUPAC) names.

2

Compound A:

Compound B:

Question 30 continues on page 21

Question 30 (continued)

- (c) Pure samples of **compound B** and **compound C** are heated to reflux with a small volume of concentrated sulfuric acid.

Draw the structural formula of the organic compound that would be produced from this reaction. **2**



- (d) Identify the name of the synthesis reaction described in part (c). **1**

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End of Question 30

Please turn over

Question 31 (6 marks)

The acid dissociation constant (K_a) of a weak acid remains constant at a set temperature, regardless of the acid's concentration, yet the *degree of ionisation (%)* changes at different acid concentrations.

The K_a of ethanoic acid is 1.75×10^{-5} at 25°C .

Use calculations to show that the addition of 50 mL of water to a 50 mL sample of 0.100 mol L^{-1} ethanoic acid solution results in a *larger* degree of ionisation (%) for the diluted acid at 25°C .

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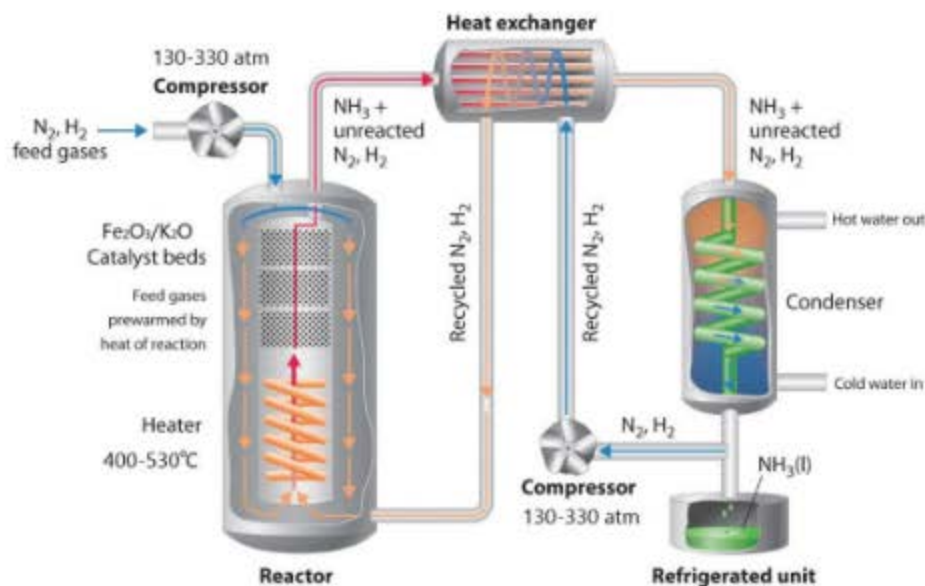
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Question 32 (5 marks)

The diagram shows the industrial process used for the production of ammonia from nitrogen and hydrogen gases.



Explain FOUR factors that may have been considered in the design of this industrial process to ensure optimal production. Make specific reference to the diagram in your response.

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Question 32 continues on page 24

Question 32 (continued)

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End of paper

Student Number: _____

Section II extra writing space. Clearly indicate which question you are answering.

Please note the use of this extra space with the question in the examination paper.

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Student Number: _____

Section II extra writing space. Clearly indicate which question you are answering.

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Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\text{Avogadro constant, } N_A \dots\dots\dots 6.022 \times 10^{23} \text{ mol}^{-1}$$

Volume of 1 mole ideal gas: at 100 kPa and

$$\text{at } 0^\circ\text{C (273.15 K)} \dots\dots\dots 22.71 \text{ L}$$

$$\text{at } 25^\circ\text{C (298.15 K)} \dots\dots\dots 24.79 \text{ L}$$

$$\text{Gas constant} \dots\dots\dots 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$\text{Ionisation constant for water at } 25^\circ\text{C (298.15 K), } K_w \dots\dots\dots 1.0 \times 10^{-14}$$

$$\text{Specific heat capacity of water} \dots\dots\dots 4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Infrared absorption data

Bond	Wavenumber/cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

¹³C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -C-C- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ R-C-Cl \text{ or } Br \\ \end{array}$	10–70
$\begin{array}{c} \\ R-C-C- \\ \quad \\ O \end{array}$	20–50
$\begin{array}{c} \\ R-C-N \\ \quad \diagup \quad \diagdown \end{array}$	25–60
$\begin{array}{c} \\ -C-O- \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \diagup \quad \diagdown \\ C=C \\ \diagdown \quad \diagup \end{array}$	90–150
$R-C \equiv N$	110–125
	110–160
$\begin{array}{c} R-C- \\ \\ O \end{array}$ esters or acids	160–185
$\begin{array}{c} R-C- \\ \\ O \end{array}$ aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	$\lambda_{\max}$ (nm)
C—H	122
C—C	135
C=C	162

Chromophore	$\lambda_{\max}$ (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for the standard potentials. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium																							
		Atomic Number Symbol Name																																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		79 Au 197.0 Gold						6 C 12.01 Carbon		9 F 19.00 Fluorine		10 Ne 20.18 Neon																					
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium								13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon															
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57-71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Astatine		85 At Radon			
87 Fr Francium		88 Ra Radium		Actinoids		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Cn Copernicium		113 Nh Nihonium		114 Fl Flerovium		115 Mc Moscovium		116 Lv Livermorium		117 Ts Tennessine		118 Og Oganesson	



**Strathfield Girls
High School**

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Student Number

HSC Chemistry Trial Examination 2022

Multiple-Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

	A	B	C	D
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

HSC Chemistry Trial Examination 2022

Marking Guidelines and Sample Answers



SECTION I (20 marks)

Multiple Choice Q1-20

Question	Answer	Question	Answer	Question	Answer	Question	Answer
1	D	6	C	11	C	16	A
2	B	7	B	12	C	17	B
3	D	8	A	13	A	18	B
4	A	9	D	14	B	19	C
5	D	10	D	15	C	20	C

SECTION II (80 marks)

Q21 (5 marks)

(a)

Criteria	Marks
- Provides an equation to show the ionisation of ethanamine in aqueous solution - Includes an equilibrium arrow and states of matter for all molecular species (both reactants)	1

Sample answer: $\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{CH}_2\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$ $\text{CH}_3\text{CH}_2\text{NH}_2$ state may be (g) or (aq)

(b)

Criteria	Marks
- Uses the $\text{p}K_b$ to calculate the K_b of ethanamine - Uses an ICE chart to indicate equilibrium concentrations - Uses the K_b expression to calculate the concentration of OH^- ions - Clearly states and applies the assumption ($0.15 - x \approx 0.15$) or uses a quadratic equation - Calculates the pOH of the solution - Calculates the pH of the solution - All working and calculations are shown	4
<i>Provides any FOUR of the following criteria with most working shown:</i> - Uses the $\text{p}K_b$ to calculate the K_b of ethanamine - Uses an ICE chart to indicate equilibrium concentrations - Uses the K_b expression to calculate the concentration of OH^- ions - Calculates the pOH of the solution - Calculates the pH of the solution	3
Provides any TWO of the above criteria (or equivalent) with some working shown	2
Provides any ONE correct and relevant calculation	1

Sample answer:

$$K_b \text{ of ethanamine} = 10^{-\text{p}K_b} = 10^{-3.29} = 5.1286 \times 10^{-4}$$

$$K_b = [\text{CH}_3\text{CH}_2\text{NH}_3^+][\text{OH}^-]/[\text{CH}_3\text{CH}_2\text{NH}_2] = 5.1286 \times 10^{-4}$$

	$[\text{CH}_3\text{CH}_2\text{NH}_2]$	$[\text{CH}_3\text{CH}_2\text{NH}_3^+]$	$[\text{OH}^-]$
I	0.15	0	0
C	-X	+X	+X
E	$0.15 - X$	X	X

As K_b is very small, we can assume: $0.15 - X \approx 0.15$

$$K_b = [\text{CH}_3\text{CH}_2\text{NH}_3^+][\text{OH}^-]/[\text{CH}_3\text{CH}_2\text{NH}_2] = X^2/0.15 = 5.1286 \times 10^{-4}$$

$$X = \sqrt{(5.1286 \times 10^{-4} \times 0.15)} = \sqrt{(7.6929 \times 10^{-5})} = 8.77 \times 10^{-3} \quad \therefore [\text{OH}^-] = 8.77 \times 10^{-3} \text{ M}$$

$$\text{pOH} = -\log(8.77 \times 10^{-3}) = 2.057$$

$$\text{pH} = 14 - \text{pOH} = 14 - 2.057 = 11.9$$

Q22 (6 marks)

(a)

Criteria	Marks
- Identifies that the equivalence point is indicated by a persistent purple colour - Explains that this is due to the excess of MnO_4^- ions as all the H_2O_2 is reacted	2
Identifies that the reaction mixture will turn purple	1

Sample answer: The equivalence point would be recognised by a persistent purple colour as the MnO_4^- titrant is added. This is caused by the first excess of MnO_4^- ions that occurs when all the H_2O_2 has reacted.

(b)

Criteria	Marks
- Calculates the average titre of acidified MnO_4^- solution (only titres within 0.1 mL) - Calculates the moles of MnO_4^- ions used in the titration - Applies the reaction mole ratio to calculate the moles of H_2O_2 reacted (in 20 mL aliquot) - Calculates the molar mass of H_2O_2 - Calculates the concentration of H_2O_2 in mol/L OR the mass of H_2O_2 in 20 mL aliquot - Calculates the concentration of H_2O_2 in g/L - All working and calculations are shown	4
<i>Provides any FOUR of the following criteria with most working shown:</i> - Calculates the average titre of acidified MnO_4^- solution (only titres within 0.1 mL) - Calculates the moles of MnO_4^- ions used in the titration - Applies the reaction mole ratio to calculate the moles of H_2O_2 reacted (in 20 mL aliquot) - Calculates the concentration of H_2O_2 in mol/L OR the mass of H_2O_2 (using molar mass) - Calculates the concentration of H_2O_2 in g/L (using molar mass)	3
Provides any THREE of the above criteria (or equivalent)	2
Provides any ONE correct and relevant calculation	1

Sample answer:

$$\text{Average titre of } \text{MnO}_4^- \text{ solution} = (26.90 + 26.80 + 26.90) \text{ mL} \div 3 = 26.867 \text{ mL (titres within 0.1 mL)}$$

$$\text{Mol } \text{MnO}_4^- = 0.0239 \text{ mol/L} \times 0.026867 \text{ L} = 6.4212 \times 10^{-4} \text{ mol}$$

$$\text{Reaction mol ratio } \text{MnO}_4^- : \text{H}_2\text{O}_2 = 2 : 5 = 1 : 5/2 \quad (\text{i.e. ratio } 2 : 5 = 2/2 : 5/2 = 1 : 5/2)$$

$$\therefore \text{mol } \text{H}_2\text{O}_2 \text{ reacted} = (5/2) \times 6.4212 \times 10^{-4} \text{ mol} = 1.6053 \times 10^{-3} \text{ mol}$$

$$\text{Molar mass } \text{H}_2\text{O}_2 = (2 \times 1.008 \text{ g/mol}) + (2 \times 16.0 \text{ g/mol}) = 34.016 \text{ g/mol}$$

$$\text{Concentration } \text{H}_2\text{O}_2 = n/V = 1.6053 \times 10^{-3} \text{ mol} / 0.020 \text{ L} = 0.0803 \text{ mol/L} \quad \text{OR}$$

$$\text{Mass } \text{H}_2\text{O}_2 \text{ in 20 mL aliquot} = n \times \text{MM} = 1.6053 \times 10^{-3} \text{ mol} \times 34.016 \text{ g/mol} = 0.0546 \text{ g}$$

Concentration H_2O_2 in g/L = mol/L $\times$ g/mol = 0.0803 mol/L $\times$ 34.016 g/mol = 2.73 g/L **OR**

Concentration H_2O_2 in g/L = mass $\div$ volume = 0.0546 g / 0.020 L = 2.73 g/L

Q23 (11 marks)

(a)

Criteria	Marks
- Identifies the anion as carbonate - Identifies the white precipitate product of step 2 as CaCO_3 - Reasons that the colourless gas must be CO_2 - Explains that CO_2 gas is produced from the reaction of a carbonate with HNO_3 in step 1	3
- Identifies the anion as carbonate - Provides some correct reasoning from ONE step (identifies CaCO_3 precipitate or CO_2 gas)	2
Identifies the anion as carbonate OR identifies CaCO_3 precipitate or CO_2 gas	1

Sample answer: The anion in the white powder is carbonate. In step 2, a white precipitate is produced when the gas is bubbled through limewater. This precipitate would be CaCO_3 and indicates a positive test for CO_2 gas. As the white powder produces CO_2 when reacted with HNO_3 (step 1), the pure white powder must contain carbonate ions.

(b)

Criteria	Marks
- Identifies the cation as barium and supports this choice with the flame test result - Explains that insoluble BaSO_4 was produced when H_2SO_4 was added in step 5 - Explains that BaCl_2 is soluble so the addition of HCl produced no precipitate in step 4 - All reasoning provided for the three tests is valid and correct	4
- Identifies the cation as barium AND <i>Any TWO of the following criteria:</i> - Supports the choice of barium with the flame test result - States that barium produces a precipitate with H_2SO_4 in step 5 - States that barium produces no precipitate with HCl in step 4	3
- Identifies the cation as barium AND provides ONE reason from the above criteria OR <i>Suggests an alternative cation that is consistent with:</i> - An insoluble sulfate being produced when H_2SO_4 was added in step 5 - A soluble chloride resulting from the addition of HCl in step 4	2
Any information relating to the question criteria	1

Sample answer: The cation in the white powder is barium. The apple green flame in the flame test is unique to barium. Further evidence is provided by the tests in steps 4 & 5. When dilute HCl was added in step 4, there was no precipitate, as the combination of barium ions with chloride is soluble. When dilute H_2SO_4 was added in step 5, a white precipitate of BaSO_4 was formed.

(c)

Criteria	Marks
- Identifies that using Na_2SO_4 would contaminate the unknown cation with Na^+ ions - Explains that the flame test may give a mixed result if Na^+ ions are also present	2
- Identifies that Na^+ ions may affect the flame test result OR - Identifies that contamination with a cation may affect subsequent tests	1

Sample answer: Using Na_2SO_4 in step 5 would contaminate the mixture with Na^+ ions. These ions may be carried through with the filtered solid in step 6, giving a mixed result in the flame test.

(d)

Criteria	Marks
- Identifies all ions that could be present in the colourless solution - Provides adequate reasoning to justify the ion mixture present - Relates to the acidic properties of nitric acid, sulfuric acid and the HSO_4^- ion	2
- Identifies some ions that could be present in the colourless solution OR - Identifies ONE ion with some justification (based on solubility or ionisation)	1

Sample answer: The ions that could be present in the colourless mixture obtained from step 6 are NO_3^- , H_3O^+ , HSO_4^- and SO_4^{2-} . These ions are present due to the addition of nitric acid at step 1 and sulfuric acid at step 6. Both nitric acid and sulfuric acid are strong acids, producing H_3O^+ ions in aqueous solution. The HSO_4^- ion is present as it is a weak acid, with only some of these ions producing SO_4^{2-} .

Q24 (7 marks)

(a)

Criteria	Marks
Writes a balanced chemical equation to represent the buffer equilibrium system	1

Sample answer: $\text{C}_2\text{H}_5\text{COOH}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{C}_2\text{H}_5\text{COO}^-_{(\text{aq})} + \text{H}_3\text{O}^+_{(\text{aq})}$ (States of matter not required for the mark)

(b)

Criteria	Marks
- States the role of a buffer is to prevent large changes in pH (or H_3O^+ concentration) - Explains how the buffer components absorb H_3O^+ and re-establish equilibrium - Explains how the buffer components neutralise base and re-establish equilibrium - Refers to the equilibrium equation in part (a) to explain how the buffer works (directions)	4
- Provides any THREE of the above criteria to describe the action of a buffer - Demonstrates a good understanding of how a buffer works	3
- Provides any TWO of the above criteria to outline the action of a buffer - Demonstrates a basic understanding of how a buffer works	2
Identifies some features of a buffer system (any relevant information)	1

Sample answer: A buffer regulates pH by maintaining the concentration of H_3O^+ ions through an equilibrium reaction where the weak acid and its conjugate base work to neutralise any excess acid and base. In the equilibrium reaction shown in part (a), if a strong acid is added, there will be an excess of hydronium ions. This will cause the reverse reaction in the equation will be favoured. The propanoate ion will react with hydronium ions to give propanoic acid and water. If a strong base is added, the hydronium ions will immediately neutralise the added base to give water. This will cause the forward reaction to be favoured to regenerate the hydronium ions consumed. Through these processes, the pH of the buffer remains at the approximately the same level despite the addition of strong acid or base.

(c)

Criteria	Marks
- The weak acid should be selected on its $\text{p}K_a$ (or K_a) as this determines the resulting pH - Prepare a buffer with equal concentrations of the weak acid and its conjugate base to ensure maximum buffering capacity	2
Provides ONE factor that is important for the preparation of buffer solution	1

Sample answer: To prepare a buffer to a certain pH, the choice of weak acid is important as the $\text{p}K_a$ of a weak acid will determine how well it ionises in solution and, consequently, the resulting pH. To ensure maximum buffering capacity, the buffer solution should have equal concentrations of the weak acid and its conjugate base.

Q25 (9 marks)

(a)

Criteria	Marks
- Writes a balanced <i>ionic</i> equation for the formation of AgCl and BaSO_4 precipitates - Includes states of matter for the two precipitates	2
<i>Provides any ONE of the following (with states for precipitates):</i> - A balanced ionic equation for the precipitation of AgCl or BaSO_4 - A balanced chemical equation for the formation of AgCl and BaSO_4 precipitates (not ionic) - A substantially correct ionic equation for the formation of AgCl and BaSO_4 (one error only)	1

Sample answer: $\text{Ba}^{2+}_{(\text{aq})} + 2\text{Cl}^{-}_{(\text{aq})} + 2\text{Ag}^{+}_{(\text{aq})} + \text{SO}_4^{2-}_{(\text{aq})} \rightarrow \text{BaSO}_{4(\text{s})} + 2\text{AgCl}_{(\text{s})}$

(b)

Criteria	Marks
- Identifies the presence of Ag^+ and SO_4^{2-} ions in solution at volume V_0 - Relates the high conductivity at volume V_0 to the presence of ions in solution - Identifies the formation of AgCl and BaSO_4 precipitates as BaCl_2 is added - States that all ions are in precipitates to explain the minimum conductivity at volume V_1	3
<i>Any THREE of the following criteria (or equivalent combination):</i> - Identifies the ions present at volume V_0 - Relates conductivity to the presence of ions in solution - Identifies that precipitation occurs as BaCl_2 is added - Relates to precipitation of ions to explain the minimum conductivity at volume V_1	2

- Identifies components at V_0 or V_1 OR - Provides some relevant information or reasoning for the high/low conductivity at V_0 or V_1	1
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Sample answer: At volume V_0 , there are Ag^+ and SO_4^{2-} ions in solution which conduct a current well, resulting in high conductivity. As BaCl_2 solution is added, precipitates of BaSO_4 and AgCl begin to form, removing ions from solution. At volume V_1 , all of the Ag^+ and SO_4^{2-} ions have formed precipitates with the Ba^{2+} and Cl^- ions, leaving virtually no ions in solution, thereby reducing the conductivity to a minimum.

(c)

Criteria	Marks
- Calculates the moles of Ag_2SO_4 present at volume V_0 - Applies the 1:1 reaction mole ratio ($\text{Ag}_2\text{SO}_4 : \text{BaCl}_2$) to determine the moles of BaCl_2 at V_1 - Calculates the volume of BaCl_2 solution added between V_0 and V_1 - All required working is shown	2
Provides ONE correct and relevant calculation OR states the volume of BaCl_2 used	1

Sample answer:

Mol Ag_2SO_4 in solution initially = $c \times V = 0.01 \text{ mol L}^{-1} \times 0.4 \text{ L} = 0.004 \text{ mol}$

Mole ratio $\text{Ag}_2\text{SO}_4 : \text{BaCl}_2$ is 1:1 $\therefore$ Mol BaCl_2 added at volume $V_1 = \text{mol Ag}_2\text{SO}_4 \text{ present} = 0.004 \text{ mol}$

Volume BaCl_2 added: $V = n / c = 0.004 \text{ mol} / 0.1 \text{ mol L}^{-1} = 0.04 \text{ L BaCl}_2$

(d)

Criteria	Marks
- Identifies the increasing excess of Ba^{2+} and Cl^- ions (excess BaCl_2) in solution after V_1 - Relates the increasing conductivity to a property of the ions in solution	2
- Identifies the excess of Ba^{2+} and/or Cl^- ions (excess BaCl_2) in solution after V_1 OR - Identifies a property of Ba^{2+} and Cl^- ions in solution that increases conductivity	1

Sample answer: After volume V_1 , there is an increasing excess of Ba^{2+} and Cl^- ions in solution. As these ions are freely moving, the level of conductivity also increases.

Q26 (4 marks)

Criteria	Marks
- Calculates the moles of H^+ available to react with the base - Calculates the moles of OH^- available to react with the acid - Calculates the excess moles of OH^- ions - Calculates the concentration of OH^- ions - Calculates the pOH of the resulting solution - Calculates the pH of the resulting solution	4
Provides any FOUR of the above criteria	3
Provides any TWO of the above criteria	2
Provides any ONE correct and relevant calculation	1

Sample answer:

$$\text{Mol H}^+ = 0.259 \text{ mol L}^{-1} \times 0.160 \text{ L} \times 2 = 0.08288 \text{ mol}$$

$$\text{Mol OH}^- = 0.695 \text{ mol L}^{-1} \times 0.185 \text{ L} = 0.1286 \text{ mol}$$

$$\text{Mol excess OH}^- = (0.1286 - 0.08288) \text{ mol} = 0.04572 \text{ mol}$$

$$\text{Concentration OH}^- = n / v = 0.04572 \text{ mol} / (0.160 + 0.185) \text{ L} = 0.1325 \text{ mol L}^{-1}$$

$$\text{pOH of solution} = -\log [\text{OH}^-] = -\log 0.1325 = 0.8778$$

$$\text{pH of solution} = 14 - \text{pOH} = 14 - 0.8778 = 13.1$$

Q27 (6 marks)**(a)**

Criteria	Marks
Identifies the acid-base property of all three salts	1

Sample answer: NaHCO_3 is basic; $\text{NH}_4\text{CH}_3\text{COO}$ is neutral; KHSO_4 is acidic.

(b)

Criteria	Marks
- Provides <u>equations</u> for the partial hydrolysis* of relevant ions in EACH salt to show effects - Explains the acid-base property of EACH salt by relating to the H_3O^+ or OH^- produced	4
<i>Demonstrates a good understanding of acidic/basic effects through the following (or equivalent):</i> - Provides <u>equations</u> for the partial hydrolysis* of TWO relevant ions to show the effects - Relates to the acid-base properties of the ions in aqueous solution OR - Describes the hydrolysis* reactions of TWO relevant ions (e.g. two flawed equations) - Explains the acid-base properties in aqueous solution by relating to H_3O^+ or OH^- produced	3
<i>Demonstrates a basic understanding of acidic/basic effects through the following (or equivalent):</i> - Provides an <u>equation</u> for the partial hydrolysis* of ONE relevant ion to show an effect OR - Describes the hydrolysis* reaction of ONE relevant ion (e.g. one flawed equation) - Relates to the acid-base property of the ion OR - Describes, in general terms, the acidic or basic effects (relating to H_3O^+ or OH^-) of TWO salts without identifying the ions responsible for these effects	2
Any information relevant to the criteria	1

* **Hydrolysis** means the equation involves water as a reactant and produces H_3O^+ or OH^-

Sample answer:

The acid-base property of each salt is caused by hydrolysis reactions of ions in aqueous solution.

The **bicarbonate ion** in NaHCO_3 is protonated by water in aqueous solution, as shown by the equation:

$\text{HCO}_3^- (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3 (\text{aq}) + \text{OH}^- (\text{aq})$. As the reaction produces a **small amount of OH^-** , this results in a **basic pH**.

The **ammonium ion** in $\text{NH}_4\text{CH}_3\text{COO}$ releases a hydrogen ion in aqueous solution, as shown by the equation:

$\text{NH}_4^+ (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{NH}_3 (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq})$. The **ethanoate ion** is protonated by water in aqueous solution, as shown by the equation: $\text{CH}_3\text{COO}^- (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{CH}_3\text{COOH} (\text{aq}) + \text{OH}^- (\text{aq})$. As **both H_3O^+ and OH^- ions** are produced in approximately equivalent amounts, this results in a **neutral pH**.

The **hydrogen sulfate ion** in KHSO_4 releases a hydrogen ion in aqueous solution, as shown by the equation:
 $\text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{SO}_4^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$. The formation of a **small amount of H_3O^+** results in an **acidic pH**.

Note: States of matter were not required in equations. Marks are not awarded for **incorrect acid/base effects** regardless of information shown in (a). The equations must show what **can occur in water**. Contradictory information (presenting both effects) was not awarded marks. Only the ammonium ethanoate salt was neutral (both effects are relevant).

(c)

Criteria	Marks
Identifies that the chemist did not prepare the salt solutions at the same concentrations	1

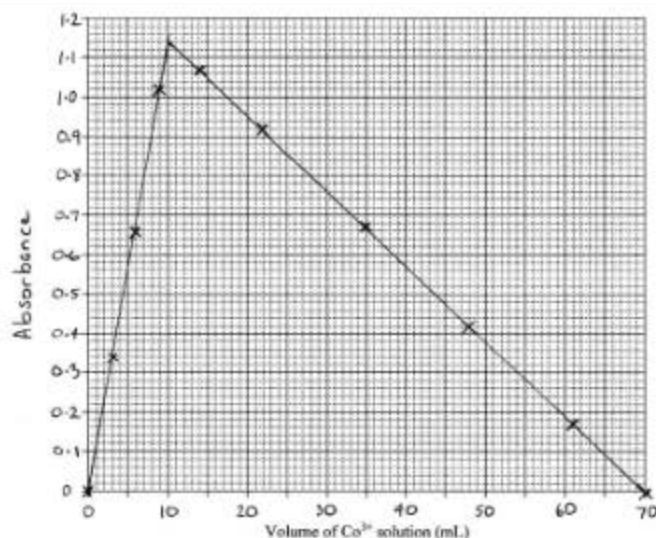
Sample answer: Equal masses of the compounds will not provide equal moles (salts have different molar masses). The salt solutions were prepared at different concentrations which means they were not fairly compared when measuring the pH of each solution.

Q28 (7 marks)

(a)

Criteria	Marks
- Labels the Y axis correctly (absorbance has no units) - Draws consistent numerical scales - Correctly plots all data points - Draws two intersecting straight lines of best fit from the X-axis through the data points	4
<i>Any THREE of the following criteria:</i> - Labels the Y axis correctly (absorbance has no units) - Draws consistent numerical scales - Correctly plots all data points - Draws two straight lines of best fit from the X-axis through the data points	3
<i>Any TWO of the following criteria:</i> - Labels the Y axis correctly (absorbance has no units) - Draws consistent numerical scales - Correctly plots all data points - Draws two straight lines of best fit from the X-axis through the data points	2
Provides a numerical scale relevant to the graph (or any combination of numeracy skills)	1

Sample answer:



(b)

Criteria	Marks
Provides an estimate of maximum absorbance based on the graph	1

Sample answer: An absorbance of 1.14 (figure shown by the graph at the intersection of the two lines).

(c)

Criteria	Marks
- Identifies the volumes of solutions required to give maximum absorbance - Relates the volumes of the reacting solutions to a reaction ratio - Identifies the correct whole number values for a and b (empirical ratio from the data)	2
Any ONE of the above criteria or equivalent information	1

Sample answer: Maximum complexation occurs at the absorbance of 1.14. As shown by the graph, the two lines of best fit intersect at a peak where 10 mL of Co^{3+} is mixed with 60 mL of ligand X solution. As these two solutions are of equal concentration, this represents a volume ratio of 10:60 and an empirical reaction ratio of 1:6 for $\text{Co}^{3+}:\text{X}$. Therefore, the values for a and b are $a = 1$ and $b = 6$.

Q29 (6 marks)

(a)

Criteria	Marks
Provides a balanced chemical equation for the neutralisation reaction	1

Sample answer: $\text{CaCO}_{3(s)} + 2\text{HCl}_{(aq)} \rightarrow \text{CaCl}_{2(aq)} + \text{CO}_{2(g)} + \text{H}_2\text{O}_{(l)}$ (States of matter not required for the mark)

(b)

Criteria	Marks
- Calculates the moles of NaOH used to neutralise the excess HCl - Calculates the moles of excess HCl (reaction with NaOH) - Calculates the moles of HCl added to the section of pipe - Calculates the moles of HCl that reacted with CaCO_3 in the pipe - Applies the reaction mole ratio to calculate the moles of CaCO_3 - Calculates the molar mass CaCO_3 - Calculates the mass of CaCO_3 that was in the pipe to 3 significant figures	5
<i>Any SIX of the following criteria:</i> - Calculates the moles of NaOH used to neutralise the excess HCl - Calculates the moles of excess HCl (reaction with NaOH) - Calculates the moles of HCl added to the section of pipe - Calculates the moles of HCl that reacted with CaCO_3 in the pipe - Applies the reaction mole ratio to calculate the moles of CaCO_3 - Calculates the molar mass CaCO_3 - Calculates the mass of CaCO_3 that was in the pipe	4
Provides any FIVE of the above criteria	3
Provides any THREE of the above criteria	2
Provides any ONE correct and relevant calculation	1

Sample answer: $\text{NaOH}_{(\text{aq})} + \text{HCl}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$

Moles of NaOH used in titration = $0.100 \text{ mol L}^{-1} \times 0.0393 \text{ L} = 0.00393 \text{ mol}$

Moles HCl excess = moles NaOH used in titration = 0.00393 mol (1:1 mole ratio)

Moles HCl added to the section of pipe = $0.250 \text{ mol L}^{-1} \times 0.100 \text{ L} = 0.0250 \text{ mol}$

Moles HCl that reacted with CaCO_3 in pipe = $(0.0250 - 0.00393) \text{ mol} = 0.02107 \text{ mol}$

Mole ratio $\text{CaCO}_3 : \text{HCl} = 1:2 \therefore \text{moles CaCO}_3 = \text{moles HCl} \div 2 = 0.02107 \text{ mol} \div 2 = 0.010535 \text{ mol}$

Molar mass $\text{CaCO}_3 = 100.09 \text{ g mol}^{-1}$

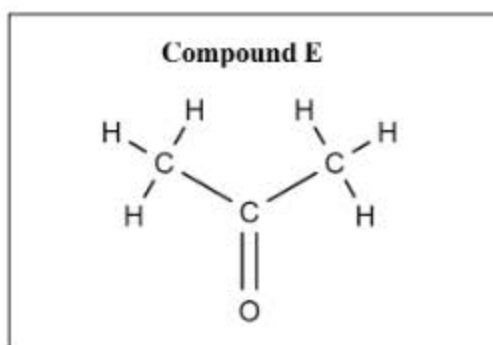
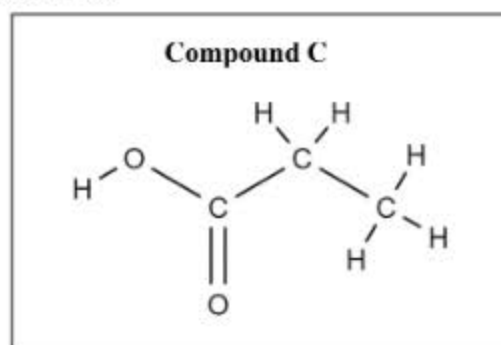
Mass $\text{CaCO}_3 = n \times \text{MM} = 0.010535 \text{ mol} \times 100.09 \text{ g mol}^{-1} = 1.05 \text{ g}$

Q30 (8 marks)

(a)

Criteria	Marks
- Provides the structures of compounds C and E - Both compounds are drawn as structural formulae (all bonds are shown)	3
- Provides the structures of compounds C and E in any format OR - Provides the structural formula of compound C or E	2
Provides the structure of compound C or E in any format	1

Sample answer:



(b)

Criteria	Marks
States the IUPAC names of compounds A and B	2
States the IUPAC name of compound A or B	1

Sample answer:

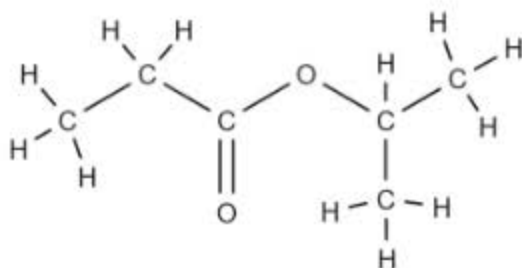
Compound A: propan-1-ol

Compound B: propan-2-ol

(c)

Criteria	Marks
- Provides the structure of the ester resulting from the reaction of compounds B and C - The structural formula is drawn (all bonds are shown)	2
Provides a substantially correct structure of the ester produced from the reaction of B and C	1

Sample answer:



(d)

Criteria	Marks
Identifies the reaction as esterification	1

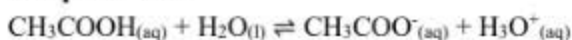
Sample answer: Esterification

Q31 (6 marks)

Criteria	Marks
- Uses the K_a of ethanoic acid and applies relevant calculations throughout the response - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.100 M ethanoic acid - Calculates the degree of ionisation (%) for 0.100 M ethanoic acid - Calculates the concentration of diluted ethanoic acid (2x dilution factor) - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.050 M ethanoic acid - Calculates the degree of ionisation (%) for 0.050 M ethanoic acid	6
- Uses the K_a of ethanoic acid and applies relevant calculations throughout the response AND <i>Provides any FOUR of the following criteria:</i> - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.100 M ethanoic acid - Calculates the degree of ionisation (%) for 0.100 M ethanoic acid - Calculates the concentration of diluted ethanoic acid (2x dilution factor) - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.050 M ethanoic acid - Calculates the degree of ionisation (%) for 0.050 M ethanoic acid	5
- Uses the K_a of ethanoic acid and applies relevant calculations throughout the response AND <i>Provides any THREE of the following criteria:</i> - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.100 M ethanoic acid - Calculates the degree of ionisation (%) for 0.100 M ethanoic acid - Calculates the concentration of diluted ethanoic acid (2x dilution factor) - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.050 M ethanoic acid - Calculates the degree of ionisation (%) for 0.050 M ethanoic acid	4
<i>Provides any THREE of the following criteria:</i> - Uses the K_a of ethanoic acid in relevant calculations - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.100 M ethanoic acid - Calculates the degree of ionisation (%) for 0.100 M ethanoic acid - Calculates the concentration of diluted ethanoic acid (2 x dilution factor) - Calculates the $[\text{CH}_3\text{COO}^-]$ at equilibrium for 0.050 M ethanoic acid - Calculates the degree of ionisation (%) for 0.050 M ethanoic acid	3
Provides any TWO of the above criteria (or equivalent information)	2

• Provides any ONE of the above criteria (or equivalent information)	1
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Sample answer:



$$K_a = [\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]/[\text{CH}_3\text{COOH}]$$

	[CH ₃ COOH]	[CH ₃ COO ⁻]	[H ₃ O ⁺]
I	0.100	0	0
C	- X	+ X	+ X
E	0.100 - X	X	X

Using ICE chart concentrations:

$$K_a = X^2/0.100 = 1.75 \times 10^{-5}$$

$$X = \sqrt{(1.75 \times 10^{-5} \times 0.100)} = 1.323 \times 10^{-3}$$

$$\therefore X = [\text{CH}_3\text{COO}^-] = \mathbf{1.323 \times 10^{-3} \text{ mol L}^{-1}}$$

$$\therefore \text{Degree of ionisation at } 0.100 \text{ mol L}^{-1} = (1.323 \times 10^{-3}/0.100) \times 100 = \mathbf{1.32\%}$$

As K_a is very small, we can assume: $0.100 - x \approx 0.100$

Ethanoic acid is diluted to 0.050 mol L^{-1} with water ($50 \text{ mL} \rightarrow 100 \text{ mL}$)

$$K_a = X^2/0.050 = 1.75 \times 10^{-5} \quad (\text{using ICE chart - same process as used previously})$$

$$X = \sqrt{(1.75 \times 10^{-5} \times 0.050)} = 9.354 \times 10^{-4}$$

$$\therefore X = [\text{CH}_3\text{COO}^-] = \mathbf{9.354 \times 10^{-4} \text{ mol L}^{-1}}$$

$$\therefore \text{Degree of ionisation at } 0.050 \text{ mol L}^{-1} = (9.354 \times 10^{-4}/0.050) \times 100 = \mathbf{1.87\%}$$

When ethanoic acid is diluted, the degree of ionisation increases.

Q32 (5 marks)

Criteria	Marks
- Describes FOUR chemical design factors that are relevant to features of the industrial process - Explains how EACH factor optimises the production of ammonia - Comprehensively relates the explanation to the diagram as relevant for each factor	5
- Explains THREE chemical design factors that optimise the production of ammonia - Makes specific reference to the diagram as relevant OR - Describes FOUR chemical design factors that optimise the production of ammonia - Makes specific reference to the diagram as relevant	4
- Explains TWO chemical design factors that optimise the production of ammonia - Makes some reference to the diagram as relevant OR - Describes THREE chemical design factors that optimise the production of ammonia - Makes some reference to the diagram as relevant	3
- Explains ONE chemical design factor that optimises the production of ammonia OR - Describes TWO chemical design factors that optimise the production of ammonia OR - Identifies FOUR chemical design factors that optimise the production of ammonia	2
Any information relevant to the above criteria	1

Sample answer:

Four factors that may have been considered in the design of this industrial process include the following:

- **Use of moderately high temperature 400-530°C.** When gases are heated, the molecules gain more kinetic energy. This results in more energetic and successful collisions and, consequently, a faster reaction rate.
- **Use of catalyst beds $\text{Fe}_2\text{O}_3/\text{K}_2\text{O}$.** Catalysts increase the reaction rate by providing an alternative reaction pathway with a lower activation energy. In the presence of the catalyst, more molecules of N_2 and H_2 have the required activation energy, so the conversion of N_2 and H_2 to the NH_3 product is faster.
- **Use of high pressure 130-330 atm.** Increasing the pressure of a gaseous reaction results in more frequent collisions between reactant molecules. This causes the reaction rate to increase as there is a greater proportion of successful collisions at high pressure.
- **Recycling of unused reactants.** Reactions are most efficient when all molecules of the reactants are converted to products. Through the repeated recycling of unused N_2 and H_2 reactant gases in the system, the complete conversion of these reactants to NH_3 product is possible. This ensures that almost 100% atom economy can be achieved. The cost of sourcing more reactants is also minimised.

The use of a moderately high temperature, along with the Fe_2O_3 catalyst and high pressure all serve to increase the reaction rate. If the reaction of N_2 and H_2 to give NH_3 is faster, the industrial process is more efficient and economic (more product in less time), as NH_3 production is optimised. The recycling of reactants until the conversion to product is complete ensures a maximum yield of NH_3 product is achieved.